

ASSESSMENT - 5

1) RELATIONAL ALGEBRA OPERATIONS (SELECTION)

```
mysql> SELECT * FROM EMPLOYEE WHERE Salary > 50000;
+-----+-----+-----+-----+
| EmpID | EmpName | DeptID | Salary |
+-----+-----+-----+-----+
| 101   | Asha    | 1      | 60000  |
| 103   | Priya   | 1      | 55000  |
| 104   | Rahul   | 2      | 70000  |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

2) RELATIONAL ALGEBRA OPERATIONS (PROJECTION)

```
mysql> SELECT EmpName FROM EMPLOYEE;
+-----+
| EmpName |
+-----+
| Asha    |
| Ravi    |
| Priya   |
| Rahul   |
| Anu     |
+-----+
5 rows in set (0.00 sec)
```

3) RELATIONAL ALGEBRA OPERATIONS (UNION)

```
mysql> (SELECT EmpName FROM EMPLOYEE WHERE DeptID = 1)
-> UNION
-> (SELECT EmpName FROM EMPLOYEE WHERE DeptID = 2);
+-----+
| EmpName |
+-----+
| Asha    |
| Priya   |
| Rahul   |
| Anu     |
| Kiran   |
+-----+
5 rows in set (0.00 sec)
```

4) RELATIONAL ALGEBRA OPERATIONS (SET DIFFERENCE)

```
mysql> SELECT EmpName FROM EMPLOYEE
-> EXCEPT
-> SELECT EmpName FROM EMPLOYEE WHERE DeptID = 1;
+-----+
| EmpName |
+-----+
| Ravi     |
| Anu     |
| Kiran    |
+-----+
3 rows in set (0.00 sec)
```

5) RELATIONAL ALGEBRA OPERATIONS(CARTESIAN PRODUCT)

```
mysql> SELECT COUNT(*) FROM EMPLOYEE, DEPARTMENT;
+-----+
| COUNT(*) |
+-----+
| 15       |
+-----+
1 row in set (0.00 sec)
```

6) RELATIONAL ALBREA OPERATIONS (JOIN)

```
mysql> SELECT e.EmpName, d.DeptName
-> FROM EMPLOYEE e
-> INNER JOIN DEPARTMENT d ON e.DeptID = d.DeptID;
```

EmpName	DeptName
Asha	IT
Ravi	HR
Priya	IT
Rahul	Finance
Anu	HR

5 rows in set (0.00 sec)

7) GROUP BY, AGGREGATE FUNCTIONS

```
mysql> SELECT Product, COUNT(*) AS TotalSales,
-> SUM(Amount) AS TotalAmount, AVG(Amount) AS AvgAmount
-> FROM SALES GROUP BY Product;
```

Product	TotalSales	TotalAmount	AvgAmount
Laptop	3	135000	45000.00
Mouse	2	12000	6000.00
Keyboard	1	6000	6000.00

3 rows in set (0.00 sec)

8) HAVING CLAUSE

```
mysql> SELECT Product, SUM(Amount) AS TotalAmount
-> FROM SALES GROUP BY Product
-> HAVING SUM(Amount) > 10000;
```

Product	TotalAmount
Laptop	135000
Mouse	12000

2 rows in set (0.00 sec)

9) CORRELATED SUBQUERY

```
mysql> SELECT EmpName, Salary
-> FROM EMPLOYEE e
-> WHERE Salary > (SELECT AVG(Salary) FROM EMPLOYEE
-> WHERE DeptID = e.DeptID);
+-----+-----+
| EmpName | Salary |
+-----+-----+
| Asha    | 60000  |
| Rahul   | 70000  |
+-----+-----+
2 rows in set (0.00 sec)
```

10) VIEW CREATION AND QUERY

```
mysql> CREATE VIEW DeptSalaryView AS
-> SELECT d.DeptName, COUNT(e.EmpID) AS TotalEmployees,
-> AVG(e.Salary) AS AverageSalary
-> FROM DEPARTMENT d JOIN EMPLOYEE e ON d.DeptID = e.DeptID
-> GROUP BY d.DeptName;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> SELECT * FROM DeptSalaryView;
+-----+-----+-----+
| DeptName | TotalEmployees | AverageSalary |
+-----+-----+-----+
| Finance  | 1              | 55000.00      |
| HR       | 2              | 47500.00      |
| IT       | 2              | 65000.00      |
+-----+-----+-----+
3 rows in set (0.00 sec)
```